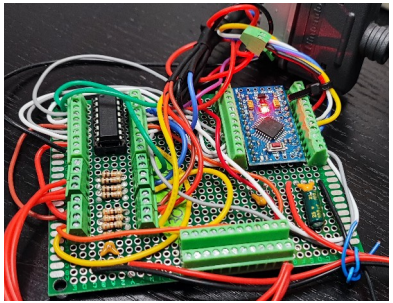
SNEngine ver. 20260409-3
USER MANUAL ver. 2

ABOUT SNENGINE

SNEngine is meant to be a high-performance, near-zero lag hardware adapter designed to connect original or third-party SNES controllers (e.g. SN30 2.4GHz) to the PC Engine console family. While designed for original wired SNES controllers, this adapter is fully compatible with the 8BitDo Retro Receiver for SNES. This allows you to use virtually any modern Bluetooth gamepad on your PC Engine (e.g. DS4 or DS5). Please note: The cases you can see on the pictures below are taken from the SNES-to-NeoGeo project. These are ideal to house SNEngine's electrical components.

Optional OLED Display



SNEngine Prototype Board

SNEngine using the SNES-to-Neo-Geo projects' case

NEW IN THIS VERSION

- **Hotkey optimization:** You can now keep SELECT held down and press UP or DOWN repeatedly to cycle through the available autofire settings and button layouts. In the previous version, both buttons had to be released before selecting the next setting.
- **Avoiding unnecessary saves:** If you press the hotkey to save your settings but nothing has changed, the settings will not be saved again.
- **OLED support:** You can now connect a small I²C OLED display to the adapter to see which settings are currently active. The display does not introduce any input lag, and voltage remains stable. It automatically turns off after a period of inactivity and turns back on when you resume playing.

Key Features

- **Dynamic Button Mapping:** Choose from 8 distinct controller layouts to match your playstyle or specific game requirements.
- **Multi-Stage Autofire (Turbo):** Integrated turbo functionality for PCE buttons I and II with 6 adjustable speed tiers, ranging from 11 Hz to 31 Hz. The mapping logic allows you to hold a dedicated standard button to charge shots instantly, even while Auto-Fire is active on the same PCE input.
- **On-the-Fly Configuration:** Switch between layouts and turbo speeds instantly using simple Hotkey combos (SELECT + D-Pad).
- **Persistent Memory:** All settings (current layout and turbo speed) can be saved to the EEPROM, ensuring your preferences are ready as soon as you power on the console.
- **Optional OLED display support**

NEXT DEVELOPMENT STAGES

I am also exploring **6-button support**. Since this requires some core hardware and software changes, it will take some more time to finalize. Thanks for following the project!

BUTTON LAYOUTS

You can choose your preferred button layout by pressing **SELECT + DOWN** simultaneously. A is the default.

Layout	SNES Buttons					
	A	B	X	Y	L	R
A	2	1	1	2	T-1	T-2
B	1	2	2	1	T-2	T-1
C	2	1	T-2	T-1	1	2
D	1	2	T-1	T-2	2	1
E	T-2	T-1	1	2	1	2
F	T-1	T-2	2	1	2	1
G	2	T-2	1	T-1	1	2
H	1	T-1	2	T-2	2	1

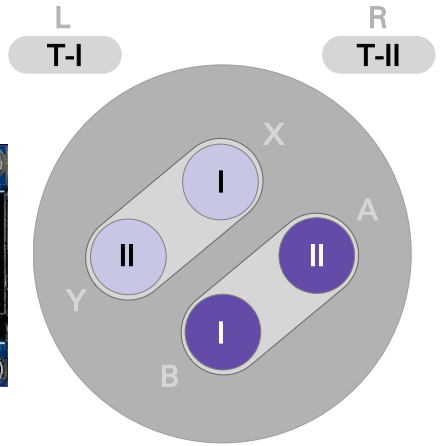
*T=Turbo

LAYOUT EXAMPLES

Assigning autofire to a shoulder button like in Layout A makes particular sense if you want to perform a charge shot with a different button. If you need autofire on both buttons anyway, you're better off using a different mapping, such as Layout H.

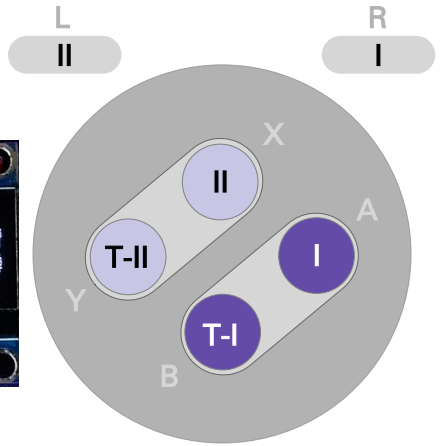
LAYOUT A

Recommended for: World Sports Competition/Power Sports, R-Type



LAYOUT H

Recommended for: Salamander



AUTOFIRE SPEED

You can adjust the autofire frequency by pressing the hotkey combination **SELECT+UP**.

- Setting 1: ~11.36 Hz (works great for Bloody Wolf)
- Setting 2: 12.50 Hz (nice for R-Type & Salamander)
- Setting 3: ~13.89 Hz
- Setting 4: ~15.63 Hz
- Setting 5: ~20.83 Hz
- Setting 6: 31.25 Hz

AUTOFIRE OVERRIDE

Signals from autofire buttons are overridden by pressing the "normal" buttons with the same number. For example, if you hold down autofire 1 and then press button 1, the signal from button 1 will take effect. The same applies to autofire 2 in conjunction with button 2. This is useful for games where you want to use both - autofire and charge shots. This kind of gameplay is really awkward to achieve with the official PC Engine controllers, since you have to activate and deactivate a special autofire toggle switch. Games where this is advantageous include:

- Detana! TwinBee,
- Cotton – Fantastic Night Dreams,
- Rayxanber II,
- R-Type I & II / R-Type Complete CD,
- Aldynes (SuperGrafx),
- 1941 Counter Attack (SuperGrafx),
- Salamander, Gradius I & II (long laser beam, not real charge).

SAVING THE SETTINGS

You can store your chosen button layout and autofire setting by pressing **SELECT+RIGHT**.

The settings will not be rewritten to the EEPROM unless they have changed since the last save.

- **Code & diagrams:** https://github.com/retrocityransom/snengine
- **Trailer:** https://youtu.be/9fGX9Tb3N7Q
- **Robin Edwards' SNES-to-NeoGeo 3D printer files:** https://github.com/robinhedwards/SNES-to-NeoGeo